RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Monday, 5 October 2026, 5:25 PM
State	Finished
Completed on	Monday, 5 October 2026, 5:36 PM
Time taken	11 mins 7 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	**g**	**t**	**a**	**b**	
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2  #include <string.h>
3  int main() {
4      char a[100], b[100];
```

```c
    int dp[101][101], i, j, m, n;
    scanf("%s", a);
    scanf("%s", b);
    m = strlen(a);
    n = strlen(b);
    for (i = 0; i <= m; i++) {
        for (j = 0; j <= n; j++) {
            if (i == 0 || j == 0)
                dp[i][j] = 0;
            else if (a[i - 1] == b[j - 1])
                dp[i][j] = dp[i - 1][j - 1] + 1;
            else
                dp[i][j] = dp[i - 1][j] > dp[i][j - 1] ?
                           dp[i - 1][j] : dp[i][j - 1];
        }
    }
    printf("%d", dp[m][n]);
    return 0;
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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